

Pyeloplasty is surgery to fix a blockage where the kidney drains into the ureter, the tube that carries urine to the bladder. If this spot is too narrow, urine backs up. That can cause pain, infections, stones or kidney damage. The surgeon removes the narrow part and makes a wide new connection.

About This Procedure

Urine collects in a funnel inside the kidney (the **renal pelvis**) and then flows into the ureter. Where they meet, a narrow spot can slow the flow. Sometimes a nearby **blood vessel** presses on it. This is called a **UPJ obstruction**. During pyeloplasty, your surgeon:

- Removes the narrow part
- Sews the healthy ureter to the kidney in a **wide funnel shape**, moving any blood vessel out of the way
- Leaves a soft **stent** (a thin tube) inside to hold it open while it heals

Most often this is done with **robotic or keyhole surgery** through a few small cuts. Sometimes one larger cut is used.

Benefits and Risks in Numbers

more than
9 in 10
repairs work and
last

about
7 in 100
adults have a
problem in the first
month, such as an
infection or leak

about
2 in 100
need another
operation in the
first month

Numbers are averages from published studies. Your own risk may differ; ask your team.

If the narrowing comes back, it can usually be fixed again.

Learn the Terms

Kidney

The organ that filters your blood and makes urine.

Ureter

The tube that carries urine from the kidney to the bladder.

UPJ

The spot where the kidney meets the ureter. This is where the blockage is.

Renal pelvis

The funnel inside the kidney where urine collects.

Stent

A soft tube inside the ureter, from kidney to bladder, that holds the repair open.

Crossing vessel

A blood vessel that can press on the UPJ and cause the blockage.

Hydronephrosis

Swelling of the kidney from urine backing up.

Cystoscopy

A look into the bladder with a thin scope. It is often how the stent is removed.

Will it hurt? You are asleep for the surgery, so you feel nothing. Afterward, the small cuts will be sore. Gas used in keyhole surgery can cause shoulder or belly aches for a day or two. The stent can cause urgency or a twinge in your side when you urinate. Many people go home the next day.

First days

Soreness and stent symptoms. Walk often and drink fluids.

Next few weeks

The stent stays in. Your team tells you when you can lift and exercise.

Stent removal

On the date your team gives you, often in the office.

After that

A kidney scan checks that urine drains well.

How to Get Ready

- Follow your instructions about eating, drinking and which medicines to stop, including blood thinners.
- A **urine test** checks for infection. Any infection is treated first.
- Arrange a ride home.

Tell your team ahead of time if you:

- Take a blood thinner
- Have had kidney or belly surgery, or a past repair of this blockage

What Happens During Surgery

- 1 You are asleep and get antibiotics.
- 2 The surgeon makes a few small cuts, or one larger cut, and finds the blocked spot.
- 3 The narrow part is removed.
- 4 The ureter is sewn to the kidney over a stent. A small drain may be left for a short time.

About Your Stent

The stent stays in for several weeks. **Keep the removal date.** A stent left in too long can crust over and block the kidney.

Call the office that did your surgery if you have:

- A temperature of **100.4 °F (38 °C)** or higher, or chills
- Side or belly pain that is getting worse
- Redness, swelling or fluid leaking from a cut

Go to an emergency department if:

- You cannot urinate
- Urine like red wine or with clots, or vomiting that will not stop
- Fever with shaking chills and feeling very ill
- You cannot reach the office and are worried

Three Things to Remember

1. Pyeloplasty removes the blockage and makes a wide new connection. It works for more than 9 in 10 people.
2. Stent symptoms are normal. Keep the date to have the stent removed.
3. Go to your follow-up scan, even if you feel well.